Pay Disparity Analysis: Base Salary by Gender, Race and Education

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1. Executive Summary

This project asks whether base salary differs by gender, race and education, and whether those differences persist after controlling for job-related factors. It uses 62,642 salary submissions (records, not unique employees). After removing 2,304 records with no base salary, 60,338 records were used in the regression models.

Key findings

- **Education shows the largest adjusted pay differences.** Compared with Bachelor's degree holders, PhD holders earn about 35.85% more and Master's holders about 15.51% more after controlling for company, job title, years of experience and years at company. Both effects are statistically significant ($p < 0.001$).
- **Controls shrink the raw gaps substantially.** The PhD gap falls from 52.78% (unadjusted) to 35.85% (adjusted). The Master's gap falls from 24.52% to 15.51%. The Some College gap falls from 19.88% to 4.60%.
- **Gender gap is small.** After controls, male submissions show base salary about 2.39% lower than female submissions ($p < 0.001$). The unadjusted gap is not significant in the log-salary model (-1.02%, $p = 0.103$).
- **Every disclosed race group earns more than Asian employees after controls.** Adjusted differences versus Asian range from +5.56% (Hispanic) to +10.97% (Two or More), with Black (+9.00%) and White (+9.82%) in between. All are significant at the 5% level.
- **Demographic data is largely missing.** Race is missing for 64.20% of records, education for 51.52% and gender for 31.83%. All demographic findings describe only the people who chose to disclose, which is the biggest limitation of this work.
- **Total compensation could not be trusted as a reconciled figure.** For 12,444 records (19.87%), reported total compensation did not equal base salary + bonus + stock grant. Base salary was therefore chosen as the outcome variable.

These are associations in self-reported data, not proof of causes or of discrimination.

2. Business Problem and Questions

Business problem. An HR or compensation team wants to know whether pay differs across demographic groups and, if so, whether the differences are explained by legitimate job-related factors such as role, company and experience, or remain unexplained.

Business questions

1. How does base salary differ by gender, race and education before controlling for any other factors?

2. Are those differences statistically significant?
3. After controlling for company, job title, years of experience and years at company, which differences remain?
4. How reliable is the data for answering these questions, and what should be fixed first?

3. Data and Tools

Item	Detail
Source	Data Science and STEM Salaries, Kaggle (Jack Ogozaly)
Size	62,642 salary submissions, 31 columns in the raw file
Unit of analysis	A salary submission (record), not necessarily a unique employee. The report uses “records” and “submissions”, not headcount
Time span	Submissions dated 2017 to 2021
Main outcome	Base salary; log-transformed base salary in the regression models
Data preparation	Excel
EDA and statistics	Python: pandas, seaborn, matplotlib, SciPy, pingouin (Welch ANOVA), statsmodels (OLS)
Dashboard	Power BI (two pages)

Workflow: business problem → data cleaning → data exploration → EDA → feature engineering → descriptive statistics → t-test and ANOVA → unadjusted and adjusted regression → dashboard → recommendations.

4. Data Cleaning and Data Quality

4.1 Column selection

The raw file had 31 columns. Eleven were dropped because they were repetitive: row_number, tag, and nine indicator columns (masters, bachelors, phd, highschool, is_asian, white, multirace, black_01, hispanic_01). The indicator columns were already summarised in the single race and education columns, so 20 columns were kept. A check of the multirace indicator confirmed that “Two Or More” records are the multiracial group.

4.2 Duplicates

No duplicate rows and no duplicate order_received values were found (0 each), so every record is a distinct submission.

4.3 Missing values

Column	Missing records	Share of 62,642
race	40,215	64.20%
education	32,272	51.52%
gender	19,941	31.83%
other_details	22,510	35.93%
level	123	0.20%
company	12	0.02%
dma_id	2	<0.01%

The high missingness in gender, race and education suggests that many people were not comfortable, or not asked, to share demographic details. Because such a large share was missing, rows were neither dropped nor imputed with a mode or other guess. Missing values were instead labelled “NA” and kept as their own category, so the analysis can show how that group behaves rather than hiding it. In total, 41,157 records (65.70%) are missing at least one of gender, race or education.

4.4 Text standardisation

The level column held 2,888 distinct raw values, many of them company-specific codes (for example Amazon L5/L6, Google L3 to L8, Microsoft 59 to 64). Text was converted to upper case and common software-engineer spellings (SDE I, SDE1, SDE-II, SDEII and similar) were mapped to SDE-1, SDE-2 and so on, which reduced distinct values to about 2,411. Because each company uses its own level ladder, there is no reliable way to map levels onto a common scale. Years of experience (YOE) was therefore used as the seniority measure instead, and level was not used in modelling.

4.5 Zero base salary records

There were 2,304 records with a base salary of 0. Of these, 2,297 had a non-zero reported total compensation but zero bonus and zero stock, which indicates the breakdown was missing rather than that base salary was truly zero. These 2,304 records (3.68%) were excluded from all base-salary analysis, leaving **60,338 records** (96.32%) for statistics and regression.

4.6 Reported total compensation did not reconcile with its components

A new column, annual_salary_calc, was created as base salary + bonus + stock grant value and compared with the reported total_yearly_compensation (renamed annual_salary). The difference was calculated as reported minus calculated.

Statistic	Value
Records checked	62,642

Statistic	Value
Records where reported ≠ calculated	12,444 (19.87%)
Mean difference	$8,792.27
Median difference	$0
25th and 75th percentile	$0
Standard deviation	about $69,933
Minimum	-$2,328,000
Maximum	$4,395,600

Four-fifths of records reconcile exactly, which is why the median is zero. The mean is positive and the tails are very large, so a small number of records drive most of the discrepancy.

What the largest differences look like. The ten largest positive differences are mostly Software Engineering Manager, Product Manager and Software Engineer records at large technology companies. In several of them the reported total equals base + bonus + stock only if the stock value is read in thousands of dollars. For example, one Facebook Product Manager record reports a total of $4,980,000 with base $380,000, bonus $200,000 and a stock value of 4,400; 380,000 + 200,000 + 4,400,000 = 4,980,000. The same pattern appears for a Facebook Software Engineer (stock 4,000), a Microsoft Engineering Manager (stock 3,750) and a Google Product Manager (stock 3,700). This is consistent with unit inconsistencies in the stock field for some submissions. Other large differences come from records where the breakdown is zero but the total is not (for example a Google Engineering Manager with total $1,600,000 and base, bonus and stock all 0), and some are negative, meaning the reported total is lower than the sum of its parts (for example total $127,000 against a calculated $137,000 in the first record).

Why total compensation and calculated components were not treated as interchangeable.

1. The two measures disagree for about one record in five, and the disagreement is not random noise: it is concentrated in high-pay roles and companies where stock is a large share of pay.
2. The differences point to at least three different causes (unit mismatch in stock, missing breakdowns, and reported totals lower than the parts), so no single correction rule would be safe.
3. A handful of extreme records (up to $4.4 million) would pull means and regression estimates around if the calculated total were used.
4. Total compensation depends on stock and bonus structures that differ widely by company and role, which would mix pay differences with equity-plan differences.

For these reasons the analysis uses base salary, which is the most consistently reported component, and does not use annual_salary_calc as a replacement for the reported total. Total compensation appears only in exploratory views, clearly labelled as reported.

4.7 Location inconsistencies

The location fields (location, city, nation, city_id, dma_id) contain inconsistencies that were documented rather than silently corrected:

- **City and country mismatch.** At least one city that is in India (Chennai) is labelled with the United States as its nation. [TO ADD: number of such mismatches, if counted.]
- **City ID collisions.** The same city_id is shared by different places, so city_id cannot be used as a reliable unique key for a location. [TO ADD: example IDs and counts, if counted.]
- **High cardinality.** There are 1,042 distinct location strings, 950 cities and 80 nations, with many small categories.

Because of these issues, location was not modelled at city level. Nation was used only in an exploratory model (see section 10.5), and any location-based conclusions should be treated as indicative.

5. Data Exploration

Role mix. The dataset is dominated by Software Engineers.

Job title	Records	Share
Software Engineer	41,231	65.82%
Product Manager	4,673	7.46%
Software Engineering Manager	3,569	5.70%
Data Scientist	2,578	4.12%
Hardware Engineer	2,200	3.51%
Product Designer	1,516	2.42%
Technical Program Manager	1,381	2.20%
Solution Architect	1,157	1.85%
Management Consultant	976	1.56%
Business Analyst	885	1.41%
Marketing	710	1.13%
Mechanical Engineer	490	0.78%
Sales	461	0.74%
Recruiter	451	0.72%
Human Resources	364	0.58%

Other structure. There are 1,068 companies, 15 job titles and 80 nations. Years of experience averages 7.20 years (median 6, maximum 69) and years at company averages 2.70 years (median 2).

Demographic composition (race, all 62,642 records).

Race	Records	Share
NA (not disclosed)	40,215	64.20%
Asian	11,772	18.79%
White	8,032	12.82%
Hispanic	1,129	1.80%
Two Or More	804	1.28%
Black	690	1.10%

Among records with a disclosed race, Asian and White submissions dominate every experience band, while Black and multiracial submissions are the least represented. Gender has three values (Female, Male, NA) and education has six (Bachelor's Degree, Highschool, Master's Degree, PhD, Some College, NA). [TO ADD: record counts for Female, Male and for each education group.]

Where the nulls are concentrated. Of the 41,157 records missing gender, race or education, 23,932 (58.15%) are Software Engineers based in the United States, followed by Product Managers (2,935) and Software Engineering Managers (2,247) in the United States. Race non-disclosure is high across all roles and experience levels: roughly 65% to 73% for Software Engineers, Product Managers, Software Engineering Managers and Data Scientists, and about 29% to 72% for smaller roles such as Recruiters, Human Resources and Business Analysts. Missing demographic data is therefore a data-collection issue across the whole file, not a problem in one small corner.

6. Feature Engineering

Feature	How it was built	Why
exp_level	YOE binned into Fresher (0 to 2), Junior (2 to 5), Mid-level (5 to 10), Senior (10 to 15) and Highly Experienced (15+)	Gives a consistent seniority measure when company levels cannot be standardised
annual_salary_calc	base_salary + bonus + stock_grant_value	Used only to test whether reported total compensation reconciles
salary_diff	annual_salary minus annual_salary_calc	Quantifies the reconciliation gap (section 4.6)
annual_salary, YOE	Renamed from total_yearly_compensation and years_of_experience	Shorter, consistent names
NA category	Missing gender, race, education and other text fields set to "NA"	Keeps 100% of eligible records and lets non-disclosure be examined as a

Feature	How it was built	Why
		group
log(base_salary)	Natural log of base salary (regression only)	Base salary is right-skewed; log makes coefficients readable as percentage differences

7. Exploratory Data Analysis

Salary by seniority and role. Median reported total compensation rises with experience in every job title. Selected examples (reported total compensation, medians):

Job title	Fresher	Junior	Mid-level	Senior
Software Engineer	$150,000	$180,000	$210,000	$240,000
Data Scientist	$155,000	$180,000	$220,000	$262,000
Product Manager	$155,000	$185,000	$224,000	$270,000
Software Engineering Manager	$159,000	$235,000	$285,000	$305,000
Business Analyst	$88,000	$115,000	$154,000	$163,500
Recruiter	$95,000	$124,500	$155,000	$197,000

This shows that job title and experience explain a large part of pay, which is why they are used as controls.

Correlations (Pearson).

Pair	Correlation
Total compensation and stock grant value	0.771
Total compensation and base salary	0.673
Total compensation and bonus	0.489
Total compensation and YOE	0.423
Base salary and YOE	0.374
Total compensation and years at company	0.166
Base salary and years at company	0.146
YOE and years at company	0.523

Years of experience is more strongly related to pay than years at the current company (0.374 against 0.146 for base salary). Total compensation moves most with stock, which is a further reason base salary is the cleaner outcome measure.

Experience curves by gender and race. Line plots of salary against YOE show tight confidence bands up to about 10 years, where most records sit, and very wide bands beyond that because few records have long experience. Differences between groups at high experience levels should not be over-read.

Trend over time (reported total compensation, median, gender disclosed). Female median minus male median, as a share of the male median:

Year	Female median	Male median	Gap
2018	$188,000	$210,000	-10.48%
2019	$188,000	$200,000	-6.00%
2020	$175,000	$186,000	-5.91%
2021	$171,000	$172,000	-0.58%

The raw gap in reported total compensation narrowed from about 10% to under 1% over four years. This view uses reported total compensation and is not controlled for role or experience, so it should be read separately from the base-salary regression results.

Race representation by seniority. Median base salary by race and experience level is similar across groups. In the unadjusted view Asian submissions do not stand out as lower paid, yet after controlling for other factors all other disclosed groups come out higher than Asian. This points to differences in workforce composition (role, company, experience mix) behind the raw picture.

8. Descriptive Statistics (Base Salary > 0)

By gender

Group	Mean	Median
Female	$136,873.97	$135,000
Male	$139,467.70	$140,000
NA	$148,333.91	$147,000

By race

Group	Mean	Median
Asian	$129,829.17	$135,000
Black	$131,549.28	$124,000
Hispanic	$132,629.76	$130,000
Two Or More	$137,812.19	$134,000
White	$139,939.49	$138,000

Group	Mean	Median
NA	$146,625.32	$145,000

By education

Group	Mean	Median
Bachelor's Degree	$123,958.50	$125,000
Highschool	$129,378.12	$129,500
Some College	$141,966.20	$138,000
Master's Degree	$145,753.48	$145,000
NA	$145,820.74	$145,000
PhD	$173,409.86	$165,000

Reading these tables. Male mean base salary is $2,593.73 (1.89%) above female. The raw gap by education is much larger: PhD holders average about $49,451 (39.9%) above Bachelor's holders. Records with an undisclosed gender, race or education have the highest averages in each table, a point that matters for interpreting every demographic result (section 13). Across all 62,642 records, the Power BI overview reports a mean base salary of $136.69K and a median of $140.00K.

9. Hypothesis Testing

Welch's tests were used because group sizes and variances are unequal. The significance threshold is 5% ($p < 0.05$).

Variable	Test	Statistic	p-value	Result
Gender (Male vs Female)	Welch t-test	t = 3.918	8.99e-05	Significant: male mean base salary is higher than female
Race (6 groups)	Welch ANOVA	F = 171.23 (df 5, 3,216.4)	6.90e-162	Significant: at least one group mean differs
Education (6 groups)	Welch ANOVA	F = 394.80 (df 5, 1,778.8)	3.27e-285	Significant: at least one group mean differs

How to read these.

- The t-statistic of 3.918 is a measure of how many standard errors apart the two means are. It is not a percentage. The actual raw gap is about $2,594, or 1.89%.
- Effect sizes are small. The partial eta-squared is 0.0145 for race (about 1.5% of salary variation) and 0.0339 for education (about 3.4%). With about 60,000 records, even small differences give very small p-values, so significance here shows that the differences are unlikely to be random, not that they are large.

- These tests compare raw averages and do not control for role, company or experience, which is why regression follows.
- ANOVA shows that some group differs but not which one, so the regression coefficients are used to find where.

10. Regression Modelling

10.1 Method

Ordinary Least Squares (OLS) with log-transformed base salary as the dependent variable, on the 60,338 records with positive base salary. In a log model a coefficient b corresponds to a percentage difference of (exp(b) - 1) × 100 versus the reference group, and that percentage is what the dashboard shows. Reference groups are Female (gender), Asian (race) and Bachelor's Degree (education). Records with undisclosed values are kept as an "NA" category so that they do not silently disappear.

- **Unadjusted models:** the demographic variable only.
- **Adjusted models:** the demographic variable + company + job title + years of experience + years at company.

10.2 Gender (reference: Female)

Group	Unadjusted coef	Unadjusted %	p	Adjusted coef	Adjusted %	p
Male	-0.0103	-1.02%	0.103	-0.0242	-2.39%	<0.001
NA	0.0731	+7.58%	<0.001	0.0302	+3.07%	<0.001

After controls, male submissions have about 2.39% lower base salary than female submissions, and the difference is statistically significant. Submissions with undisclosed gender are about 3.07% higher than female. The unadjusted gap in the log model is not significant, even though the raw-dollar Welch t-test is. This happens because the t-test works on raw salaries (more sensitive to extreme values) while the regression works on log salaries; the result also suggests that role mix and experience partly hide a small gap that appears once they are held constant.

10.3 Race (reference: Asian)

Group	Unadjusted coef	Unadjusted %	Adjusted coef	Adjusted %	Adjusted p
Black	0.0763	+7.93%	0.0862	+9.00%	<0.001
Hispanic	0.0774	+8.05%	0.0541	+5.56%	<0.001
Two or More	0.1309	+13.99%	0.1041	+10.97%	<0.001
White	0.1339	+14.33%	0.0937	+9.82%	<0.001
NA	0.1855	+20.38%	0.1095	+11.57%	<0.001

All unadjusted and adjusted p-values are below 0.001. Every disclosed race group earns more than Asian submissions even after controls. The largest adjusted effect among disclosed groups is Two or More

(+10.97%), which is the figure shown on the dashboard. Black and Hispanic groups are small (690 and 1,129 records), so their estimates are less precise.

10.4 Education (reference: Bachelor's Degree)

Group	Unadjusted coef	Unadjusted %	p	Adjusted coef	Adjusted %	p
Highschool	0.0296	+3.00%	0.264	0.0366	+3.73%	0.075
Some College	0.1813	+19.88%	<0.001	0.0450	+4.60%	0.022
Master's Degree	0.2193	+24.52%	<0.001	0.1442	+15.51%	<0.001
PhD	0.4238	+52.78%	<0.001	0.3064	+35.85%	<0.001
NA	0.2192	+24.51%	<0.001	0.1304	+13.93%	<0.001

Master's and PhD holders have significantly higher base salary than Bachelor's holders, and the premium survives controls. Highschool is not significant at the 5% level in either model (p = 0.075 after adjustment). The fall in coefficients after adjustment shows that job allocation, experience and company explain part of the raw education premium.

10.5 Model fit and checks

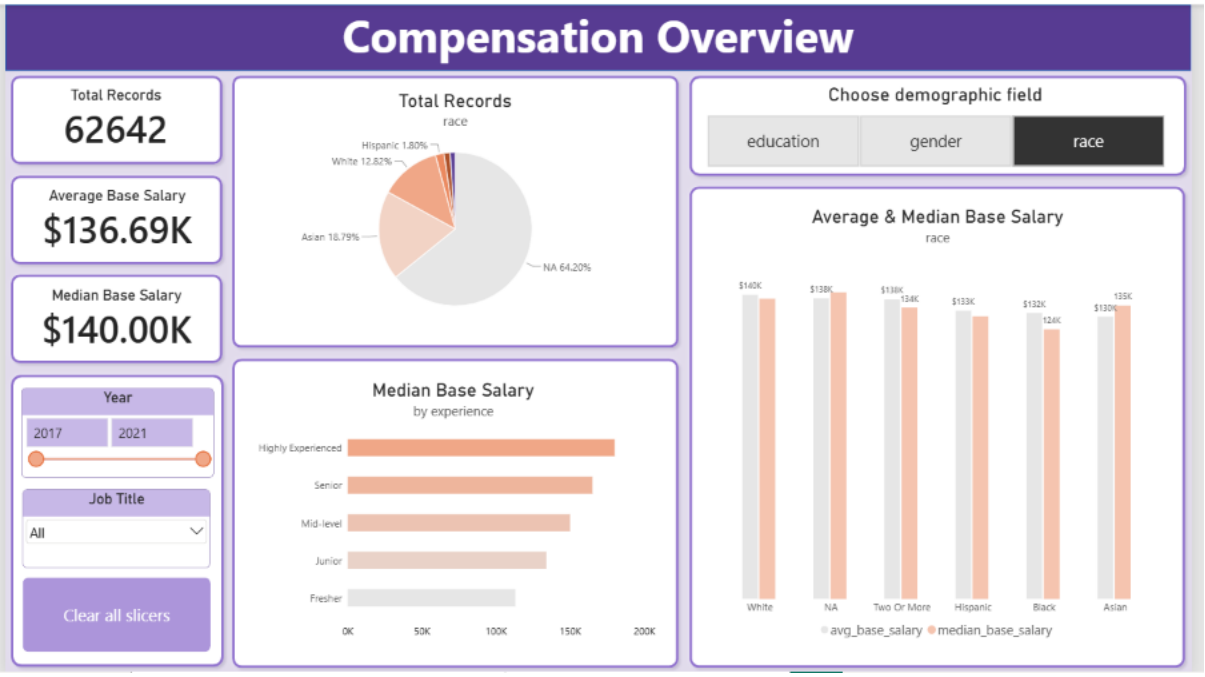
Model	Unadjusted R^2	Adjusted model R^2	Adjusted model Adj. R^2
Gender	0.006	0.443	0.433
Race	0.023	0.448	0.438
Education	0.042	0.457	0.447

- **Fit.** Demographics alone explain 0.6% to 4.2% of the variation in log base salary. Adding company, title and experience raises this to about 44% to 46%, so most of the explainable pay variation comes from job-related factors, not demographics.
- **Heteroscedasticity.** Because variance is not equal across records, a robust (HC3) version of the gender model was run. The gender coefficient was unchanged (-0.0242) and remained significant ($t = -5.14$, $p < 0.001$). The HC3 runs for the race and education models were prepared but not executed in the notebook, so those two models rely on standard OLS errors for now.
- **Residuals.** Residuals from the unadjusted gender model show skewness of -1.77 and kurtosis of 8.69, so they are not normal. With this sample size, coefficient estimates are still informative, but p-values near the threshold (such as Highschool at 0.075 and Some College at 0.022) should be interpreted with care.
- **Location control.** Company and nation overlap heavily and produced a singular-matrix warning when entered together, so the adjusted demographic models control for company but not nation. An exploratory model using nation + YOE + years at company + job title (no demographics) reached $R^2 = 0.682$, which shows that location explains a large share of pay that the demographic models do not capture.

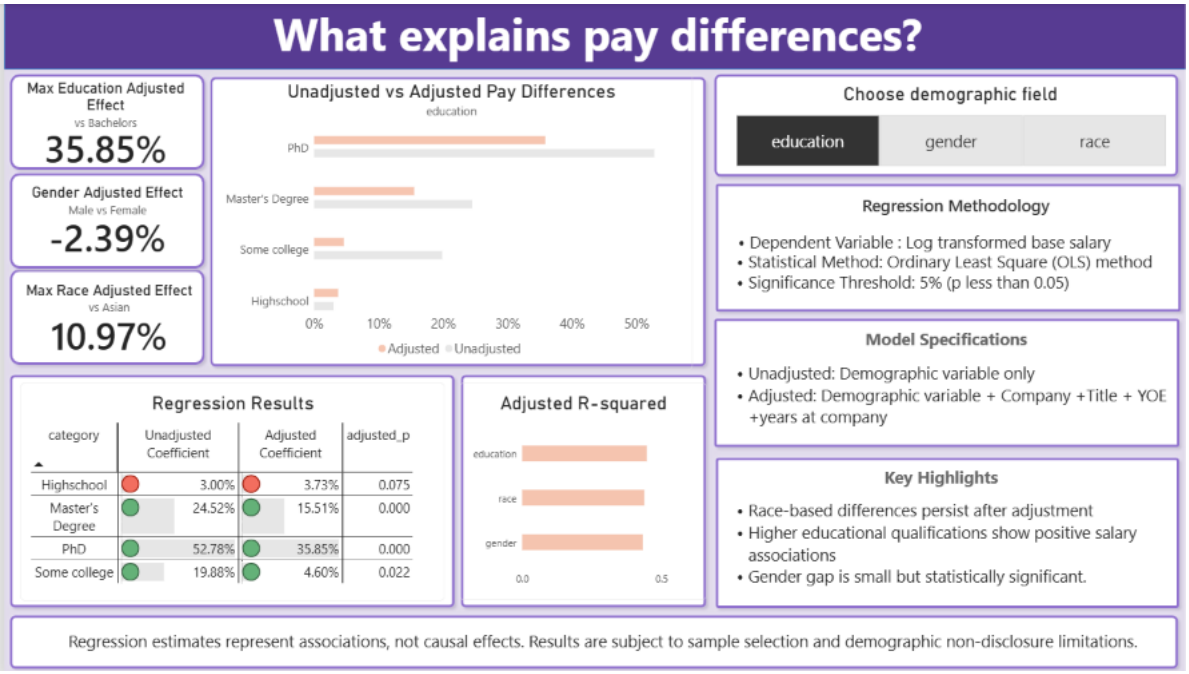
11. Dashboard (Power BI)

The findings were turned into a two-page interactive Power BI dashboard.

Page 1, Compensation Overview. Cards show total records (62,642), average base salary ($136.69K) and median base salary ($140.00K). A demographic selector (education, gender, race) drives a records-by-group chart and an average-versus-median base salary chart. Slicers filter by year (2017 to 2021) and job title, and a chart shows median base salary by experience level. The page exists to show the data's shape, including the large share of undisclosed race.



Page 2, What explains pay differences? Cards show the largest adjusted effect for education (35.85%, vs Bachelor's), gender (-2.39%, Male vs Female) and race (10.97%, vs Asian). A chart compares unadjusted and adjusted differences for the selected demographic field, and a regression table lists unadjusted and adjusted effects with adjusted p-values. A methodology panel, an adjusted R-squared view, key highlights and a caveat that results are associations, not causal effects, complete the page.



Design decisions. Effects are shown as percentages with the reference group stated; the selector switches all visuals together; and methodology and limitations are on the page so that viewers can see how the numbers were produced.

12. Business Recommendations

1. **Fix demographic data collection first.** With 64.20% of race, 51.52% of education and 31.83% of gender values missing, no pay-equity conclusion can be considered complete. Make the demographic fields part of the standard intake form, separate “prefer not to say” from “not collected”, and explain how the data will be used so that people are more willing to disclose.
2. **Standardise compensation reporting.** Define base, bonus and stock as separate, unit-checked fields, add validation rules (for example flag stock values that look like thousands or totals that differ from the sum of their parts by more than a set tolerance), and record the currency and period. This would remove the 19.87% reconciliation gap and allow total compensation to be analysed.
3. **Run adjusted pay reviews, not raw comparisons.** Raw gaps are misleading: the PhD gap drops from 52.78% to 35.85% and the Some College gap from 19.88% to 4.60% once role, company and experience are considered. Regular reviews should compare people in comparable roles, company and experience bands.
4. **Investigate the race pattern behind the Asian reference group.** After controls, all other disclosed groups earn 5.56% to 10.97% more than Asian submissions. Asian employees are the largest disclosed group (18.79% of records), so this deserves a look at leveling, role placement and promotion patterns before any compensation action. Check whether the result holds with better location and level data before acting on it.

5. **Monitor the small gender gap.** The adjusted gap is -2.39% for male submissions, with significance but a modest size. Track it over time with the same controls and re-test it once demographic coverage improves.
6. **Review how degrees are rewarded.** The Master's (15.51%) and PhD (35.85%) premiums persist after controls. Decide whether pay bands should reflect degrees directly or only through level and role, and whether the premium is consistent across job families.
7. **Improve representation data and outreach.** Black (690), Hispanic (1,129) and multiracial (804) records are each 1.10% to 1.80% of the data. Small groups give uncertain estimates, so collecting more data from these groups would improve future analysis.
8. **Operationalise the dashboard.** Refresh it on a regular schedule, keep the caveat visible and use it as a screening tool that points reviewers to groups needing a closer, case-by-case check.

13. Limitations

- **Heavy missing data.** Race (64.20%), education (51.52%) and gender (31.83%) are largely missing. Records with undisclosed values have the highest pay in every demographic view (mean base salary of $148,333.91 for undisclosed gender against $136,873.97 for female), so non-disclosure is not random and disclosed records may not represent everyone.
- **Concentrated sample.** 65.82% of records are Software Engineers, and 58.15% of records with missing demographics are United States Software Engineers. Findings may not apply to other roles or countries.
- **Self-reported, voluntary submissions.** Values cannot be verified, and the data may reflect who chooses to submit.
- **Records, not people.** The unit is a submission, so one person could appear more than once and headcount cannot be inferred.
- **Association, not causation.** The regression shows differences that remain after controlling for the listed factors; it cannot show why they exist or establish discrimination.
- **Omitted and weak controls.** Company-specific levels could not be standardised, location is not controlled in the demographic models, and factors such as performance, skills, negotiation and education field are not in the data.
- **Company and nation overlap.** The two cannot be entered together without a singular-matrix warning, so one had to be left out.
- **Total compensation reliability.** Reported totals disagree with their parts for 19.87% of records, so only base salary was modelled.
- **Location data quality.** City-country mismatches and city ID collisions limit location analysis.
- **Small groups.** Black, Hispanic and multiracial groups have fewer than 1,200 records each, so their estimates are less stable.
- **Statistical assumptions.** Residuals are non-normal and variance is unequal. HC3 checks were completed for the gender model only.
- **Outliers.** Extreme values (such as 69 years of experience, and very large totals) were identified but not formally trimmed.
- **Time.** All records are from 2017 to 2021, so they may not reflect current pay.

14. Future Scope

1. **Treat missingness formally.** Compare disclosers and non-disclosers on role, company and experience, and use multiple imputation or inverse-probability weighting as sensitivity checks.
2. **Strengthen the models.** Add location (nation or city) with company fixed effects, test interactions such as gender × title or race × experience, and use clustered or robust standard errors for every model.
3. **Complete the robustness checks.** Run HC3 for the race and education models, check influential points, and test the results with and without extreme values.
4. **Test intersectional effects.** Examine combinations of gender, race and education instead of one variable at a time.
5. **Repair the data.** Correct the city-country mismatches and city IDs, detect unit errors in the stock field and rebuild a reconciled total compensation measure so that bonus and equity can be analysed.
6. **Compare outcomes.** Repeat the analysis for total compensation, bonus and stock once reconciled, and look at the gender trend over time (the raw reported-total gap narrowed from -10.48% in 2018 to -0.58% in 2021).
7. **Extend the dashboard.** Add confidence intervals, a time view, drill-through by company and role, and automated refresh.
8. **Validate externally.** Compare with other salary datasets or internal HR data to see whether the patterns hold.

Appendix: Items to Confirm Before Sharing

- Record counts for Female and Male, and for each education group, to complete section 5.
- Counts for city-country mismatches and the city ID collisions (section 4.7).
- Whether the Power BI overview averages ($136.69K mean, $140.00K median) exclude the 2,304 zero-base-salary records.
- Master's Degree mean base salary: the notebook gives $145,753.48 and the results workbook gives $145,744.01; this report uses the notebook figure.